

Course Code: BEM- C302
Course Name: Engineering Mathematics-III

MM: 100 Time: 3 Hr. L T P 3 1 0	Sessional examination : 30 End Semester Examination: 70 Credit : 4
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Prerequisites:	Engineering Mathematics I, Engineering Mathematics II
Objectives:	This course provides an introduction to the basic concepts and techniques of: 1. Laplace transform and its application to the solution of ordinary differential equations. 2. Fourier transform and its application to solve partial differential equations. 3. Z transform of elementary sequences both from the definition and by using tables and use the appropriate theorems to calculate Z transforms and inverse Z transforms. 4. Basic theory of function of a complex variable and theory of contour integration using residue calculus. 5. Errors and numerical solution of algebraic and transcendental equations.
Course Coordinator	Dr Lokesh Kumar Joshi
Course Faculty	Dr. Lokesh Kumar Joshi, Dr Vivek Goel
Lectures	40 Hours

NOTE:	The question paper shall consist of two sections (Section-A and Section-B). Section-A shall contain ten (10) short answer type questions of six (06) mark each and student shall be required to attempt any five (05) questions. Section-B shall contain eight (08) long answer type questions of ten (10) marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus
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UNIT	Module	Course Content	No. of Hours	POs mapped
UNIT-1	Module-1	Laplace Transform: Definition, Laplace transform of elementary functions, Shifting theorems, Transform of derivatives, Differentiation and Integration of transforms,	04	PO1/PO2/PO3/PO4/PO5/PO6/PO7,PO09,PO11/PO12
	Module-2	Heaviside unit step and Dirac Delta functions, Convolution theorem,	02	
	Module-3	Solution of ordinary linear differential equations used in Mechanics, Electric circuits and Bending of beams.	02	
UNIT-2	Module-3	Fourier Transform: Definition of Fourier transform, Fourier sine and cosine transforms. Fourier integral formula, Parseval's identity	06	PO1/PO2/PO3/PO4/PO5/PO6/PO7,PO09,PO11/PO12
	Module-4	Applications of Fourier transform in solving heat equations.	02	
UNIT-3	Module-5	Z Transform: Definition, Linearity property, Z transform of elementary functions, Shifting theorems, Initial and final value theorem, Convolution theorem,	04	PO1/PO2/PO3/PO4/PO5/PO6/PO7,PO09,PO11/PO12
	Module-6	Inversion of Z transforms, Solution of difference equations by Z transforms.	03	
UNIT-4	Module-7	Function of Complex Variable: Definition, Limit and Continuity of functions of Complex Variables: Analytic Functions, Harmonic Conjugate, Cauchy-Riemann Equations (without proof), Line Integral, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof).	05	PO1/PO2/PO3/PO4/PO5/PO6/PO7,PO09,PO11/PO12

	Module-8	Singular Points, Poles & Residues, Residue Theorem, Application of Residues theorem for Evaluation of Real Integral (UnitCircle).	04	
UNIT-5	Module-9	Errors and Roots of equations: Absolute, relative, round-off and truncation errors. Significant digits. Algebraic and Transcendental Equations, Numerical solution, Method of bisection, Newton-Raphson method, Direct iterative method, convergence.	08	PO1/PO2/PO3/PO4/PO5/PO6/PO7,PO09,PO11/PO12
Total No. of Hours			40	

Course Outcome:	On completion of this course, the students will be able to 1. apply Laplace transform in various engineering problems and solve the differential equations arising in mechanics and electrical circuits(L2, L3, L5, L6) 2. understand the concept of Fourier transform and use it to solve partial differential equations having initial and boundary values(L1, L2, L3, L4). 3. apply Z transform to convert discrete-time signals to the Z-domain, analyze system behavior, and use these techniques in digital signal processing and control systems. (L1, L2, L3, L4). 4. learn the functions of complex variables and apply it to solve the problems of complex differentiation and integration(L1, L2, L3, L5). 5. solve algebraic and transcendental equations by applying iterative methods and analyze their convergence.(L2,L3, L4,L5)
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Suggested books:

S. No.	Name of Authors /Books /Publisher	Year of Publication
1.	Kreyszig E., Advanced Engineering Mathematics 10 edition, Wiley India Pvt. Ltd, ISBN-9788126554232	2015
2.	Gerald, C.F., Wheatley P.O., Applied Numerical Analysis, Pearson, 2007, ISBN-ISBN-10 032119019X	2007
3.	Grewal B.S., Higher Engineering Mathematics, Khanna, New Delhi, ISBN-9788174091956	2000
4.	Jain R. K., Iyenger S.R.K., Jain M.K., Numerical Methods for Scientific and Engineering Computation, New Age International Publishers, ISBN- 812242001X	2010

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Scheme of Evaluation

- ❖ Attendance required: 75 %
- ❖ End semester exam: 70 marks (complete syllabus)
- ❖ Sessional Exam: 20 marks
- ❖ Assignment/seminar/tutorial: 10 marks(Each student in small groups will apply these concepts to solve practical problems)